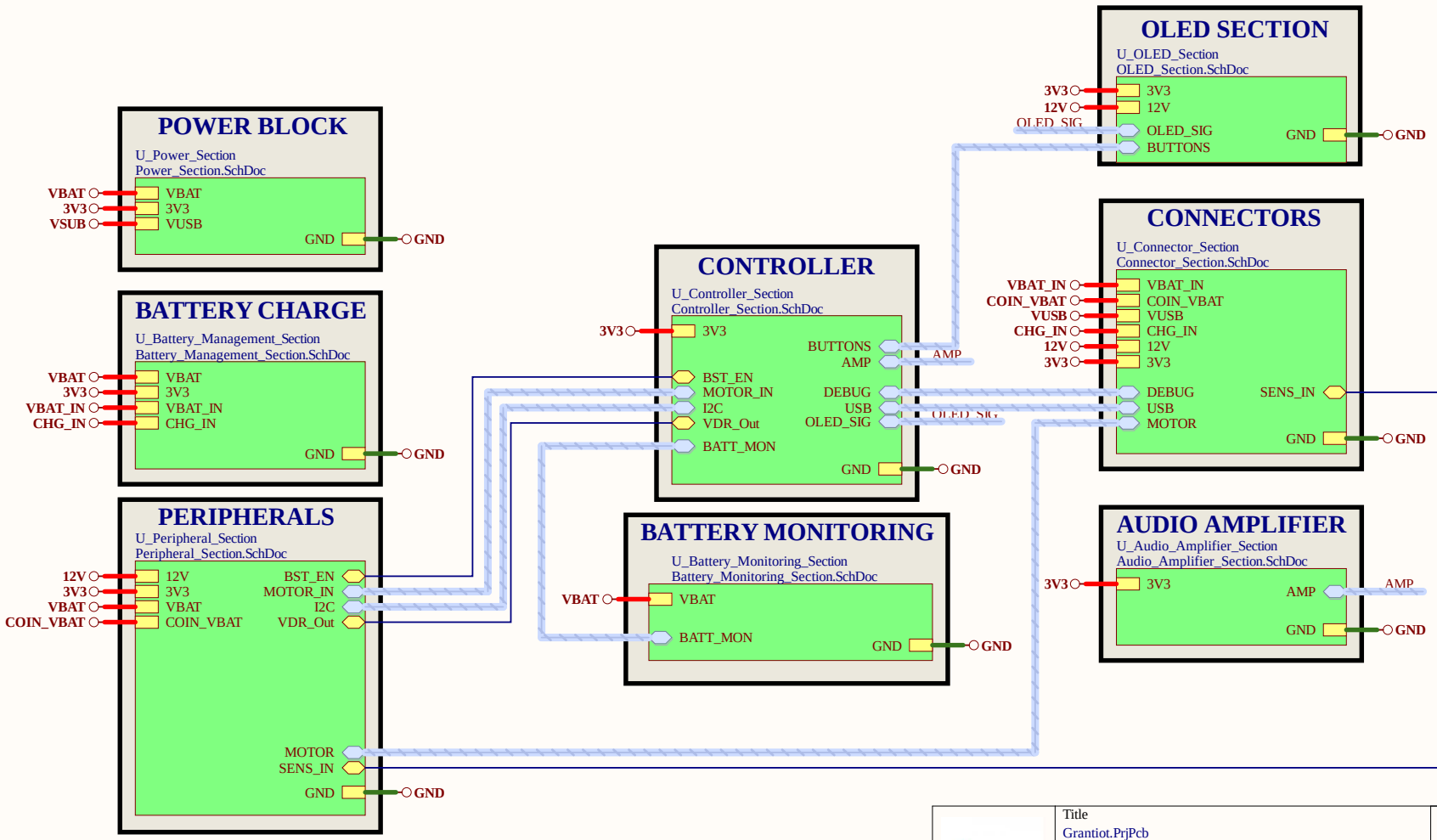
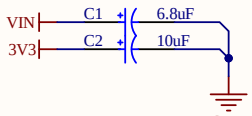
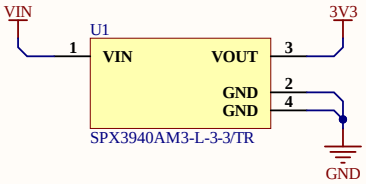


Title	Grantiot.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	3/17/2022	Sheet1 of	9
File:	D:\eSpark\..\Top Sheet.SchDoc	Drawn By:	Huzaifa Ali

DECOUPLING CAPS



MCU VOLTAGE REGULATOR IC



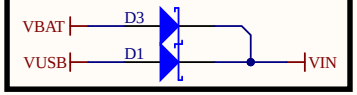
TEST POINT



POWER INDICATION LED



ORING CIRCUIT

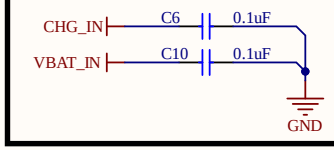


VBAT=6V
VUSB=5V
VF of Diode= 360mV @1A
VR=30V
IF=1A

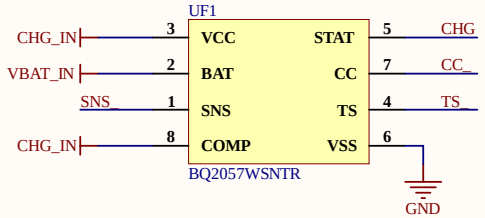


Title	Grantiot.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	3/17/2022	Sheet2 of	9
File:	D:\eSpark\..\Power_Section.SchDoc	Drawn By:	Huzaifa Ali

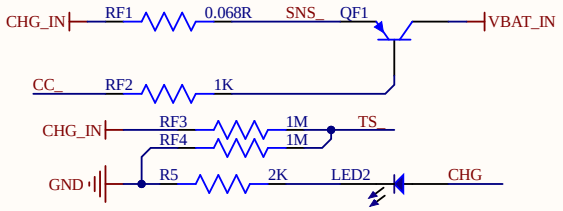
DECOUPLING CAPS



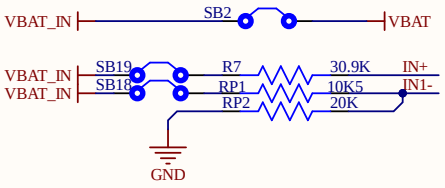
BATTERY CHARGING IC



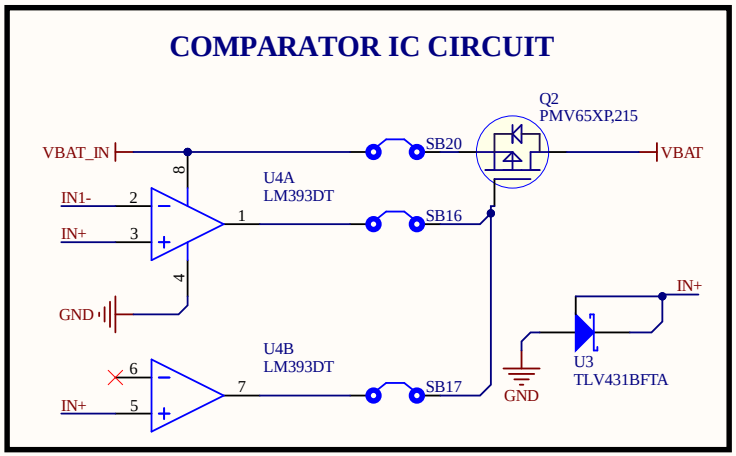
BATTERY CHARGING IC CIRCUIT



Divider for Battery Cutoff at 3.7V (3xAA Cell)



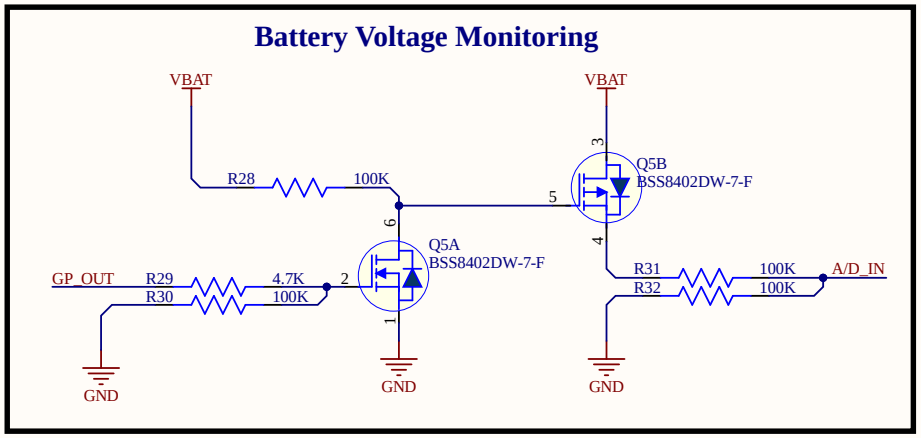
COMPARATOR IC CIRCUIT



CHG_IN	CHG_IN
VBAT_IN	VBAT_IN
VBAT	VBAT
3V3	3V3
GND	GND

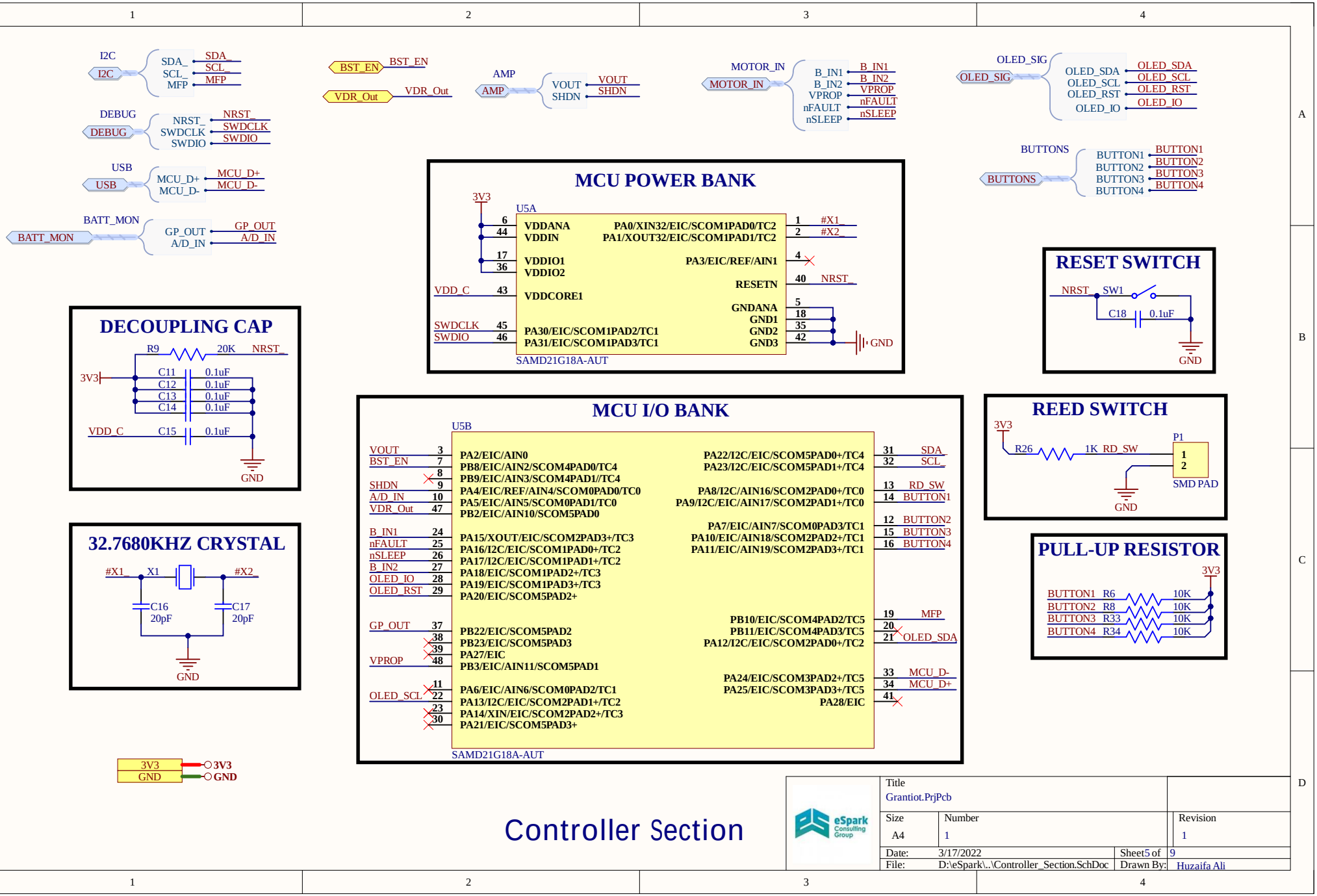
Title	Grantiot.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	3/17/2022	Sheet 3 of	9
File:	D:\eSpark\..\Battery_Management_Section.SchDoc	Drawn By:	Huzaifa Ali





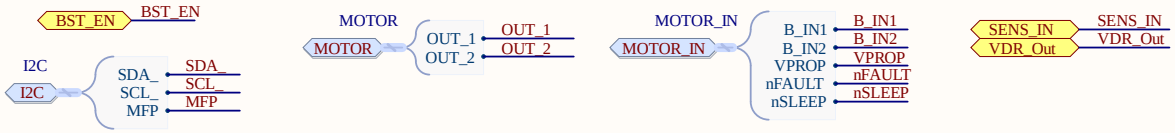


Title: Grantiot.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 3/17/2022		Sheet 4 of 9	
File: D:\eSpark\..\Battery_Monitoring_Section.SchDoc		Drawn By: Huzaifa Ali	

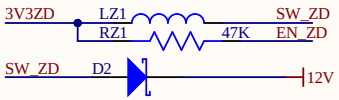


Controller Section

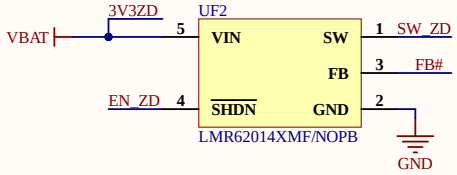
Title	Grantiot.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	3/17/2022	Sheet5 of	9
File:	D:\eSpark\..\Controller_Section.SchDoc	Drawn By:	Huzaifa Ali



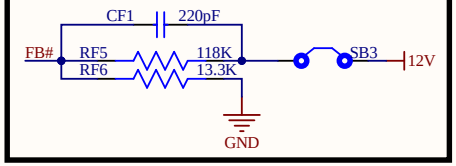
SUPPORTING CIRCUIT



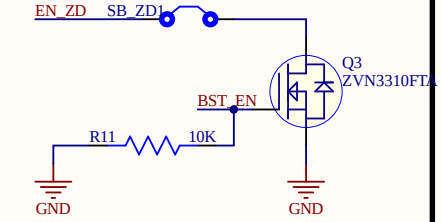
12V BOOST REGULATOR



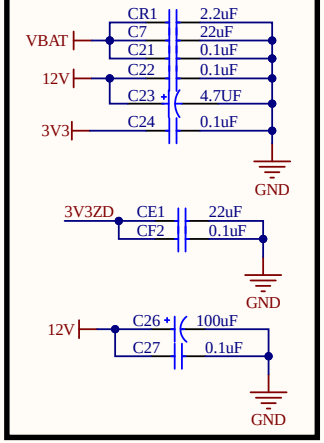
FEEDBACK NETWORK



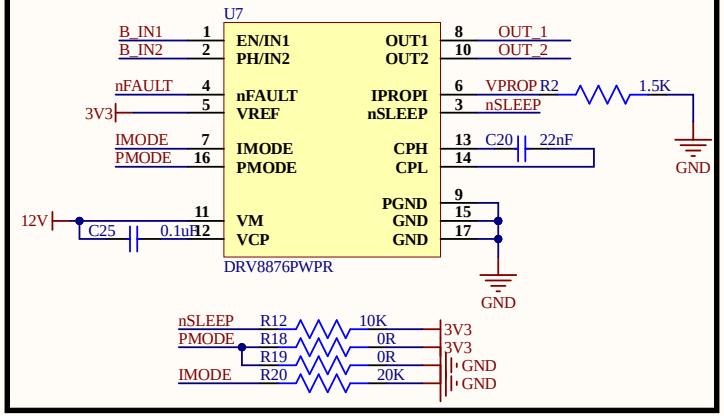
FEEDBACK NETWORK



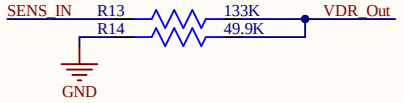
DECOUPLING CAPS



MOTOR DRIVER IC



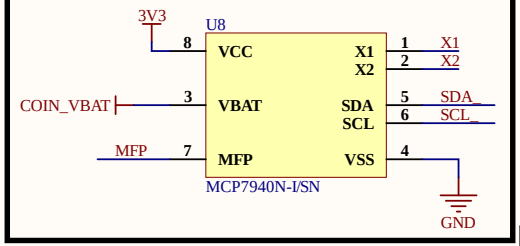
SENSOR INPUT CONVERSION



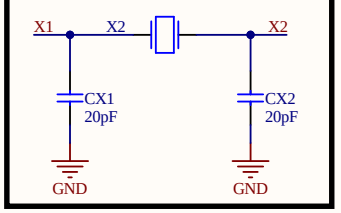
I2C PULL RESISTORS

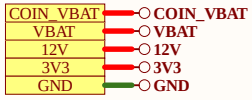


RTC CLK/CALENDAR IC



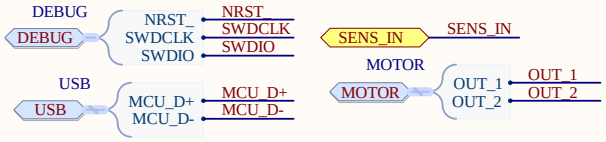
CRYSTAL 32.7680KHZ



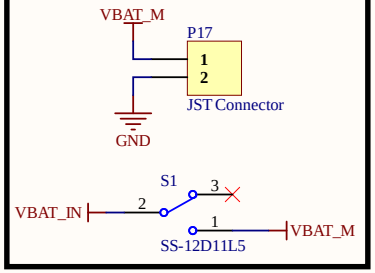


Peripheral Section

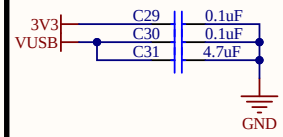
Title: Grantiot.PrjPcb		
Size: A4	Number: 1	Revision: 1
Date: 3/17/2022	Sheet 6 of 9	
File: D:\eSpark\..\Peripheral_Section.SchDoc	Drawn By: Huzaifa Ali	



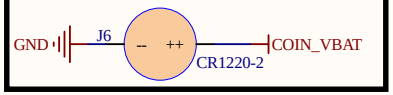
BATTERY INPUT



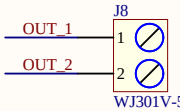
DECOUPLING CAP



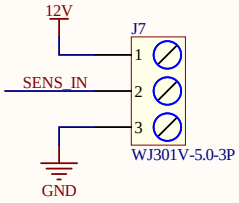
COIN CELL CONNECTOR



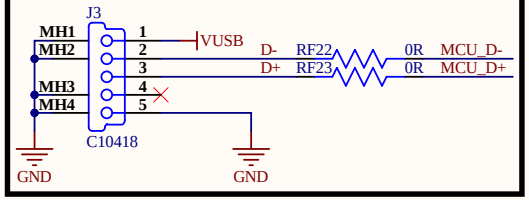
MOTOR OUTPUT



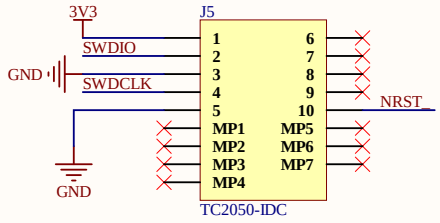
PROXIMITY SENSOR



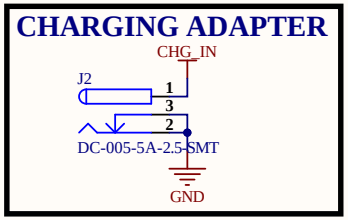
USB DATA TRANSFER



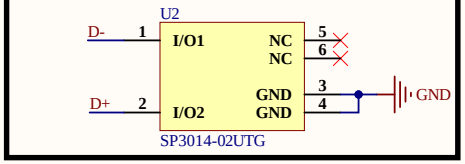
JTAG DEBUGGING CONNECTOR



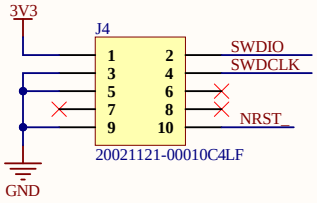
CHARGING ADAPTER



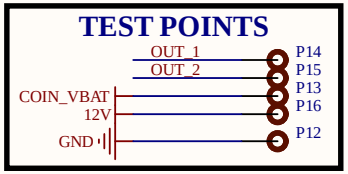
TVS ARRAY FOR USB INPUT

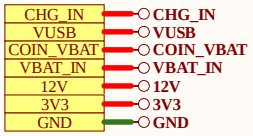


SERIAL WIRE DEBUGGING HEADER



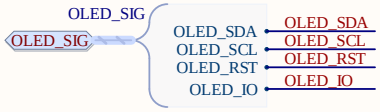
TEST POINTS

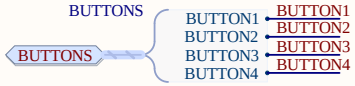


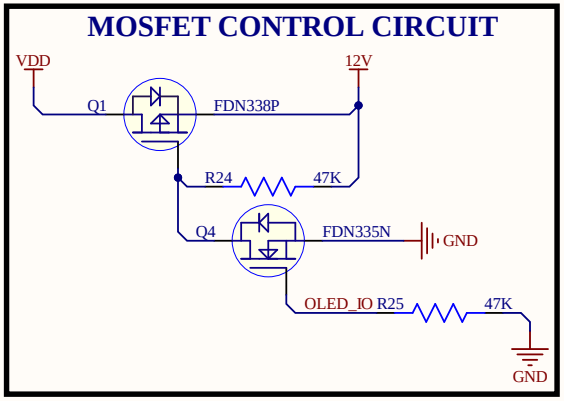


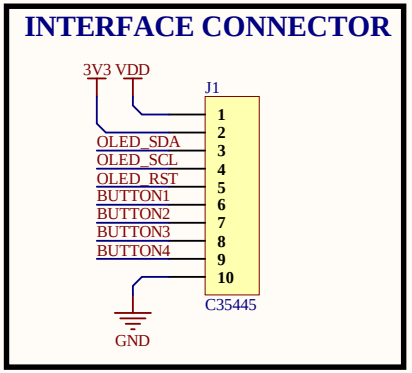
Connector Section

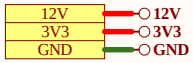
Title: Grantiot.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 3/17/2022		Sheet 7 of 9	
File: D:\eSpark\..\Connector_Section.SchDoc		Drawn By: Huzaifa Ali	











Title: Grantiot.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 3/17/2022		Sheet 8 of 9	
File: D:\eSpark\..\OLED_Section.SchDoc		Drawn By: Huzaifa Ali	

AMP

AMP

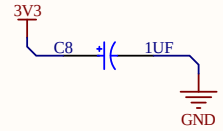
VOUT

SHDN

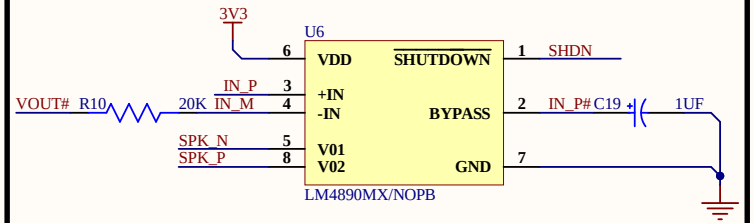
VOUT

SHDN

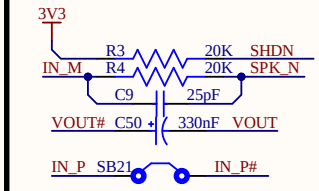
DECOUPLING CAP



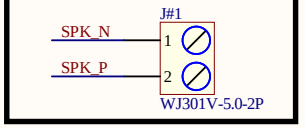
1W AUDIO AMPLIFIER IC



SUPPORTING CIRCUIT



SPEAKER OUTPUT





Title: Grantiot.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 3/17/2022		Sheet 9 of 9	
File: D:\eSpark\..\Audio_Amplifier_Section.SchDoc		Drawn By: Huzaifa Ali	